Document: JVET-AI0076-v2

Joint Video Experts Team (JVET)
of ITU-T SG 16 WP 3 and ISO/IEC JTC 1/SC 29
35th Meeting, Sapporo, JP, 12–19 July 2024

Title:	**AHG12: The Modification of BV Candidate List Construction**
Status:	Input document to JVET
Purpose:	Proposal

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Abstract

ECM-13.0 searches the block vectors of neighboring blocks in BV candidate list construction, which are derived from both adjacent spatial candidate blocks and non-adjacent spatial candidate blocks. This contribution proposes two methods to modify the adjacent spatial candidate blocks in the process of BV candidate list construction. Firstly, we replaced conventional adjacent spatial candidate blocks with the adjacent spatial candidate blocks used in OBIC mode. Secondly, we simplfied adjacent spatial candidate blocks of OBIC mode.. Experimental results on top of ECM-13.0 are reported as below:

Method (1) AI: { -0.02% Y, 0.00% U, -0.06% V, 100.5% EncT, 101.6% DecT} for Overall

{ -0.07% Y, 0.05% U, 0.00% V, 100.8% EncT, 101.4% DecT} for class F

{ -0.10% Y, -0.11% U, -0.10% V, 100.4% EncT, 101.6% DecT} for class TGM

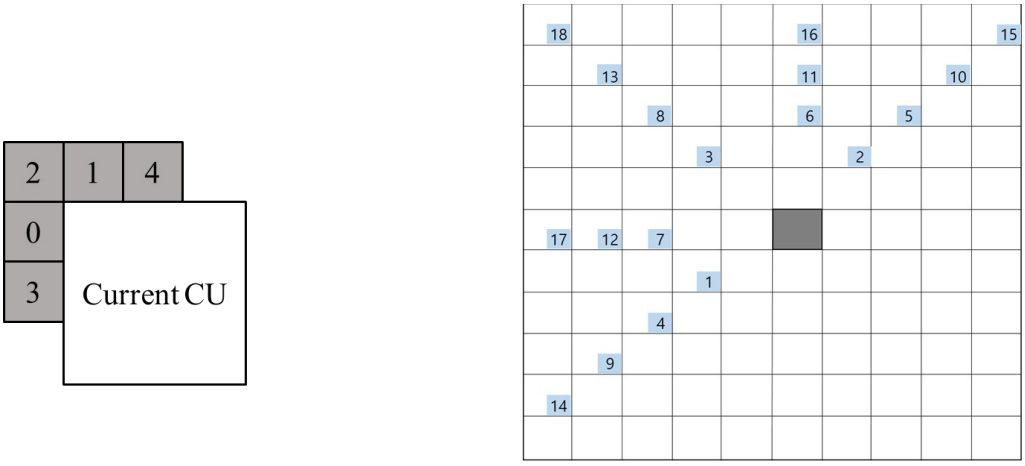
Method (2) AI: { -0.01% Y, -0.01% U, 0.01% V, 100.2% EncT, 100.5% DecT} for Overall

{ -0.11% Y, 0.05% U, 0.00% V, 100.6% EncT, 100.4% DecT} for class F

{ -0.08% Y, -0.11% U, -0.10% V, 100.5% EncT, 100.5% DecT} for class TGM

1 Introduction

In ECM-13.0 [1], the BV candidate list is constructed in DIMD, OBIC, SGPM, and IntraTMP [2]. The BV candidate list is composed of BVs derived from conventional adjacent and non-adjacent spatial candidate blocks. As shown in Fig. 1, there are five adjacent spatial candidate blocks and eighteen non-adjacent spatial candidate blocks in ECM-13.0.



(a) Conventional adjacent spatial candidate blocks (b) Non-adjacent spatial candidate blocks

Figure 1. Spatial candidate blocks used in BV candidate list construction.

2 Proposed Method

In this contribution, we propose two methods to modify the adjacent spatial candidate blocks in the process of BV candidate list construction. Firstly, we replaced conventional adjacent spatial candidate blocks with the adjacent spatial candidate blocks used in OBIC mode as shown in Fig. 2. In other words, Method 1 extended the number of adjacent spatial candidate blocks in the same manner as OBIC mode. Secondly, we modified the adjacent spatial candidate blocks of Method 1 without exceptional cases of Fig. 2 (b) in Method 2.

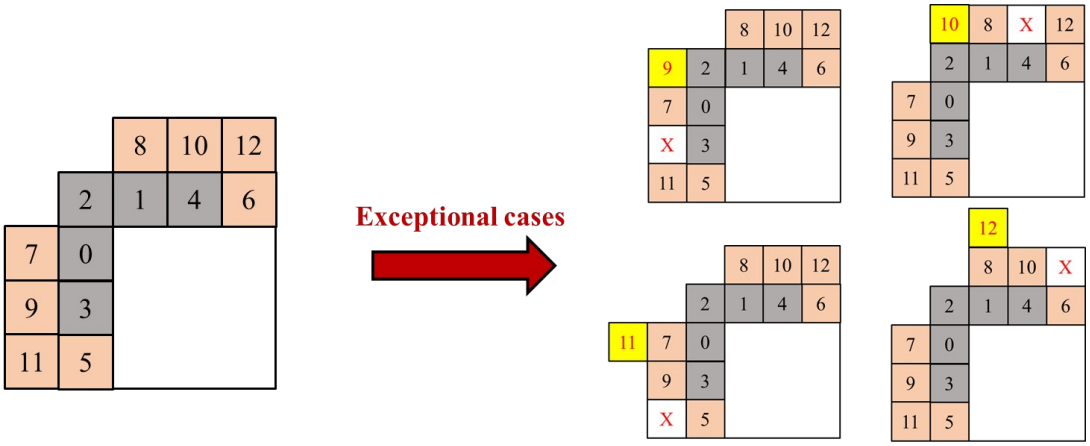


Figure 2. Adjacent spatial candidate blocks used in OBIC mode.

3 Experimental Results

The table below summarizes the coding efficiency, runtime, and peak memory consumption of the proposed methods under the JVET common test conditions [3].

Table 1. Performance of the Method 1 on top of ECM-13.0 (AI)

	All Intra Main 10						
	Over ECM-13.0						
	Y	U	V	EncT	DecT	EncVmPeak	DecVmPeak
Class A1	-0.01%	0.00%	0.03%	100.2%	101.8%	101.3%	100.1%
Class A2	-0.01%	0.01%	-0.02%	100.4%	101.7%	98.4%	100.1%
Class B	-0.01%	-0.06%	-0.14%	100.5%	101.6%	103.0%	100.0%

Class C	-0.01%	0.00%	0.03%	100.6%	101.4%	100.0%	100.0%
Class E	-0.03%	0.08%	-0.15%	100.6%	101.5%	101.0%	100.0%
Overall	-0.02%	0.00%	-0.06%	100.5%	101.6%	100.9%	100.0%
Class D	0.01%	0.06%	-0.03%	100.9%	101.7%	100.0%	100.0%
Class F	-0.07%	0.03%	0.01%	100.8%	101.4%	99.5%	100.0%
Class TGM	-0.10%	-0.07%	-0.07%	100.4%	101.6%	101.3%	100.2%

Table 2. Performance of the Method 2 on top of ECM-13.0 (AI)

	All Intra Main 10						
	Over ECM-13.0						
	Y	U	V	EncT	DecT	EncVmPeak	DecVmPeak
Class A1	0.00%	-0.01%	-0.04%	100.3%	100.3%	99.7%	99.9%
Class A2	-0.01%	-0.06%	-0.01%	100.0%	100.4%	98.9%	100.0%
Class B	0.00%	0.00%	0.01%	100.3%	100.6%	103.0%	100.0%
Class C	0.00%	0.02%	0.09%	100.1%	100.4%	99.5%	100.0%
Class E	-0.03%	-0.02%	0.00%	100.2%	100.6%	103.2%	99.9%
Overall	-0.01%	-0.01%	0.01%	100.2%	100.5%	101.0%	100.0%
Class D	0.01%	0.06%	-0.19%	100.4%	100.3%	100.1%	100.0%
Class F	**-0.11%**	0.05%	0.00%	100.6%	100.4%	99.5%	99.9%
Class TGM	**-0.08%**	-0.11%	-0.10%	100.5%	100.5%	101.2%	100.0%

4 Conclusions

The proposed method was implemented on top of ECM-13.0 with the modified BV candidate list construction. The first method replaced conventional adjacent spatial candidate blocks with the adjacent spatial candidate blocks used in OBIC mode. The second method modified the adjacent spatial candidate blocks of Method 1 without exceptional cases. Experimental results show that the proposed method (2) provides 0.01%, 0.11%, and 0.08% average luma BD-rate gain in Overall, Class F, and Class TGM, respectively. It is suggested that Method 2 can be studied in the next round of EE2.

5 References

[1] ECM-13.0 reference software, https://vcgit.hhi.fraunhofer.de/ecm/ECM.git

[2] M. Coban et al., “Algorithm description of Enhanced Compression Model 13 (ECM 13)”, JVET-AH-2025, Apr. 2024.

[3] M. Karczewicz, Y. Ye, “Common Test Conditions and evaluation procedures for enhanced compression tool testing”, JVET-Y2017, Jan. 2022.

6 Patent rights declaration(s)

DONG-A University may have current or pending patent rights relating to the technology described in this contribution and, conditioned on reciprocity, is prepared to grant licenses under reasonable and non-discriminatory terms as necessary for implementation of the resulting ITU-T Recommendation | ISO/IEC International Standard (per box 2 of the ITU-T/ITU-R/ISO/IEC patent statement and licensing declaration form).